Chapter 3

Neuromodulation: How Chemical Context Shapes Learning and Behavior

> *"Like the entomologist in pursuit of brightly colored butterflies, my attention hunted, in the flower garden of the gray matter, cells with delicate and elegant forms, the mysterious butterflies of the soul, the beating of whose wings may some day (who knows?) clarify the secret of mental life."*
>
> *— Santiago Ramón y Cajal, Recollections of My Life (1937)*

Why Neurons Alone Are Not Enough

Chapter 2 built a neuron. It receives inputs on its dendrites, integrates them, and when the membrane at the axon initial segment crosses threshold it fires an action potential that travels down the axon and crosses synapses onto other cells. That account is accurate, and it explains how signals move through the brain in milliseconds. It leaves out something that any account of learning needs: a reason why the same neuron, carrying the same signal, should change its connections on one occasion and not on another.

If neural function were set entirely by fast electrical signaling through fixed wiring, the brain would be a complex but rigid machine. The same input would reliably produce the same output, and whatever learning occurred would occur whenever neurons fired, with no regard for whether the moment deserved it. That is not what organisms do. A sound can be ignored at one moment and trigger rapid learning at the next. A mistake can be shrugged off today and permanently reshape behavior tomorrow. Even when the same neurons fire in similar patterns, learning sometimes happens and sometimes does not, and the difference lies in the state of the animal rather than in the pattern.

Electrical signaling explains how neurons communicate. It does not explain when learning happens, what is worth learning, or why the organism's internal state matters so much. Answering those questions requires looking at the chemical environment in which neural activity unfolds, and that environment is set by a small number of cell groups whose job is not to carry information but to change how everything else handles it.[1] This is neuromodulation, and it is the subject of this chapter.

Fast Signals and Slow Control

Neural communication runs on more than one timescale. At the fastest, an action potential crosses a synapse in under a millisecond, and the glutamate or GABA it releases acts on receptors a few tens of nanometers away and is cleared within milliseconds. Figure 3.1A shows this arrangement: one sender, one receiver, a cleft about 20 nanometers wide. Fast transmission of this kind carries perception, reflexes, and moment-to-moment decisions.

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[1] Marder, E. (2012). Neuromodulation of neuronal circuits: Back to the future. *Neuron*, 76(1), 1–11; Dayan, P. (2012). Twenty-five lessons from computational neuromodulation. *Neuron*, 76(1), 240–256.

Learning cannot run at that speed, and should not. If every fluctuation of activity produced a lasting change, the brain would overwrite what it knew as fast as it acquired it; this is the stability-plasticity dilemma, and Chapter 5 returns to it.[2] A learning system needs a way to let some experiences leave traces while most do not, and the decision has to be made on the basis of something slower and broader than the activity of any one synapse.

Neuromodulators are that something. Dopamine, norepinephrine, serotonin, and acetylcholine are released by small groups of neurons whose axons branch across large territories of the brain, and much of their release is not into a synaptic cleft but from swellings along the axon, called varicosities, into the extracellular space (Figure 3.1B). From there the molecule diffuses over micrometers and reaches receptors on many cells and many synapses at once; this is called volume transmission, in contrast to the point-to-point transmission of Figure 3.1A.[3] The signals themselves can be fast. A dopamine neuron's burst in response to a surprise lasts about a tenth of a second, and locus coeruleus neurons answer a salient stimulus with a similar latency. What is slow is the action. Most neuromodulator receptors are not ion channels but G-protein-coupled receptors, which work by setting off chains of chemical reactions inside the cell, and their effects on excitability and synaptic strength unfold over seconds to minutes and sometimes longer.[4]

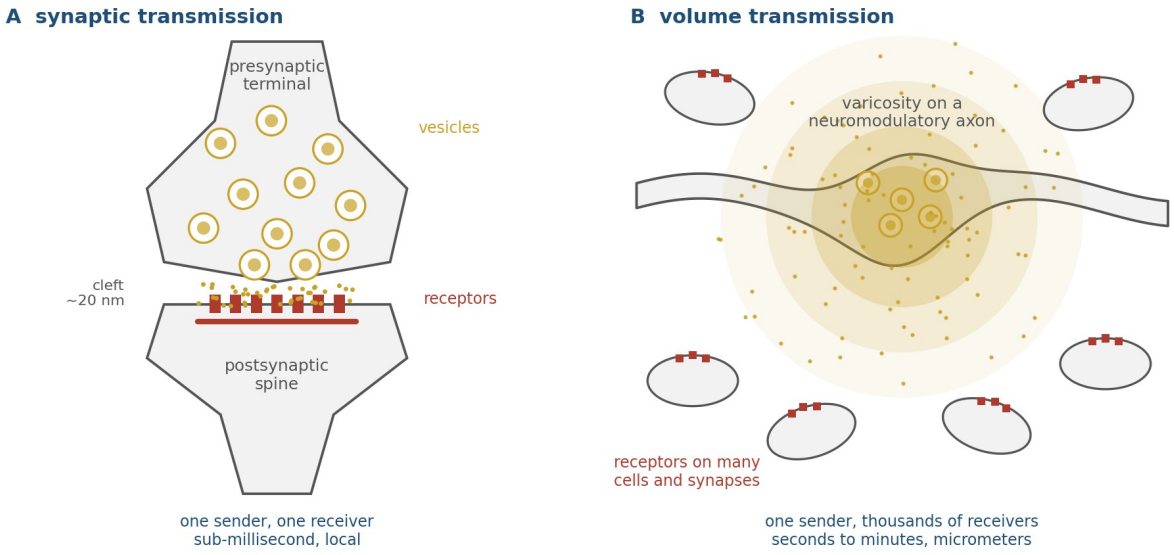


Figure 3.1. Synaptic transmission (A) versus volume transmission (B). A fast transmitter crosses a cleft about 20 nanometers wide to receptors on one partner cell. A neuromodulator released from an axonal varicosity diffuses through the extracellular space and reaches receptors on many cells and synapses, with effects that last seconds to minutes.

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[2] Carpenter, G. A., & Grossberg, S. (1988). The ART of adaptive pattern recognition by a self-organizing neural network. *Computer*, 21(3), 77–88; Abraham, W. C., & Robins, A. (2005). Memory retention: The synaptic stability versus plasticity dilemma. *Trends in Neurosciences*, 28(2), 73–78.

[3] Fuxe, K., Dahlström, A. B., Jonsson, G., Marcellino, D., Guescini, M., Dam, M., Manger, P., & Agnati, L. (2010). The discovery of central monoamine neurons gave volume transmission to the wired brain. *Progress in Neurobiology*, 90(2), 82–100.

[4] Nadim, F., & Bucher, D. (2014). Neuromodulation of neurons and synapses. *Current Opinion in Neurobiology*, 29, 48–56; Schultz, W. (1998). Predictive reward signal of dopamine neurons. *Journal of Neurophysiology*, 80(1), 1–27; Sara, S. J. (2009). The locus coeruleus and noradrenergic modulation of cognition. *Nature Reviews Neuroscience*, 10(3), 211–223. The receptor-level cascades are slow; the phasic dopamine and locus coeruleus responses that trigger them are not.

The consequence is that a neuromodulator does not carry detailed sensory content. It changes how circuits respond to the content they are already receiving: it alters excitability, biases the competition between pathways, and decides whether, how strongly, and in which direction synapses are permitted to change.[5] Figure 3.2 shows the principle at a single synapse. The same three presynaptic spikes produce a small response when the modulator level is low and a larger one when it is high, and at the high level the synapse is also opened to lasting change.

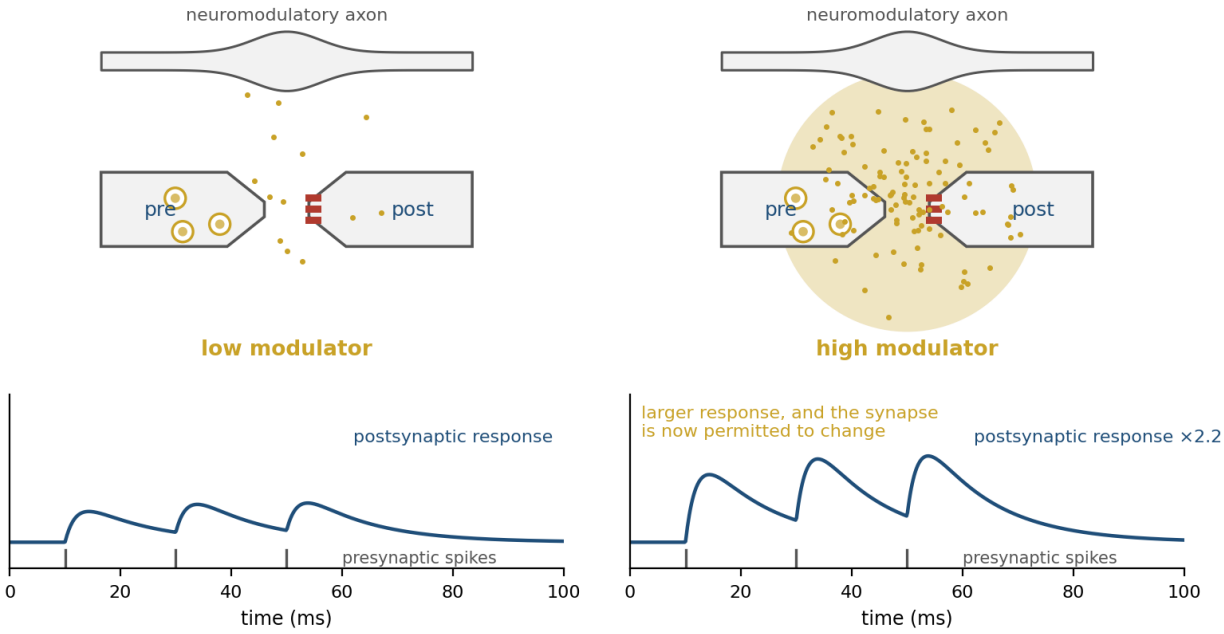


Figure 3.2. Neuromodulators create chemical context at synapses. The same synapse receiving the same three presynaptic spikes gives a small response when the modulator level is low (left) and a larger response when it is high (right); the traces are illustrative. High modulator also opens the synapse to lasting change.

It is worth being precise about what "opened to lasting change" means, because the rest of the book depends on it. Synaptic plasticity, the lasting change in the strength of a connection, can be induced by presynaptic and postsynaptic activity alone: the potentiation and depression that Chapter 5 describes are produced in brain slices by patterns of stimulation with no neuromodulator added. What neuromodulators do is gate and scale much of that plasticity in the intact brain, and decide whether it is consolidated. Kittens whose visual cortex was depleted of both norepinephrine and acetylcholine failed to show the rewiring that normally follows closing one eye. Adult rats in which a tone was paired with stimulation of the cholinergic nucleus basalis, or of the dopaminergic ventral tegmental area, expanded the area of auditory cortex devoted to that tone, something the tone alone never does.[6] Hebbian plasticity is the mechanism; neuromodulation is, for much of it, the permission.

[5] Bargmann, C. I. (2012). Beyond the connectome: How neuromodulators shape neural circuits. *BioEssays*, 34(6), 458–465.

[6] Bear, M. F., & Singer, W. (1986). Modulation of visual cortical plasticity by acetylcholine and noradrenaline. *Nature*, 320(6058), 172–176; Kilgard, M. P., & Merzenich, M. M. (1998). Cortical map reorganization enabled by nucleus basalis activity. *Science*, 279(5357), 1714–1718; Bao, S., Chan, V. T., & Merzenich, M. M. (2001). Cortical remodelling induced by activity of ventral tegmental dopamine neurons. *Nature*, 412(6842), 79–83.

Why Neuromodulatory Systems Are Diffuse

The neurons that release these modulators are few, and they sit far from the regions they control. In 1964 Annica Dahlström and Kjell Fuxe, using a method that made monoamines fluoresce, mapped the dopamine, norepinephrine, and serotonin cell groups of the rat brainstem; the cholinergic groups of the basal forebrain and brainstem were charted two decades later by Marsel Mesulam and colleagues.[7] Figure 3.3 shows the four systems on a schematic midsagittal view. Dopamine neurons sit in the ventral tegmental area and substantia nigra of the midbrain and project to the striatum and the prefrontal cortex. Norepinephrine comes almost entirely from the locus coeruleus, a nucleus in the pons that sends axons to nearly the whole forebrain and to the cerebellum. Serotonin comes from the raphe nuclei along the midline of the brainstem. Acetylcholine comes from the basal forebrain and from two nuclei in the brainstem that reach the thalamus.

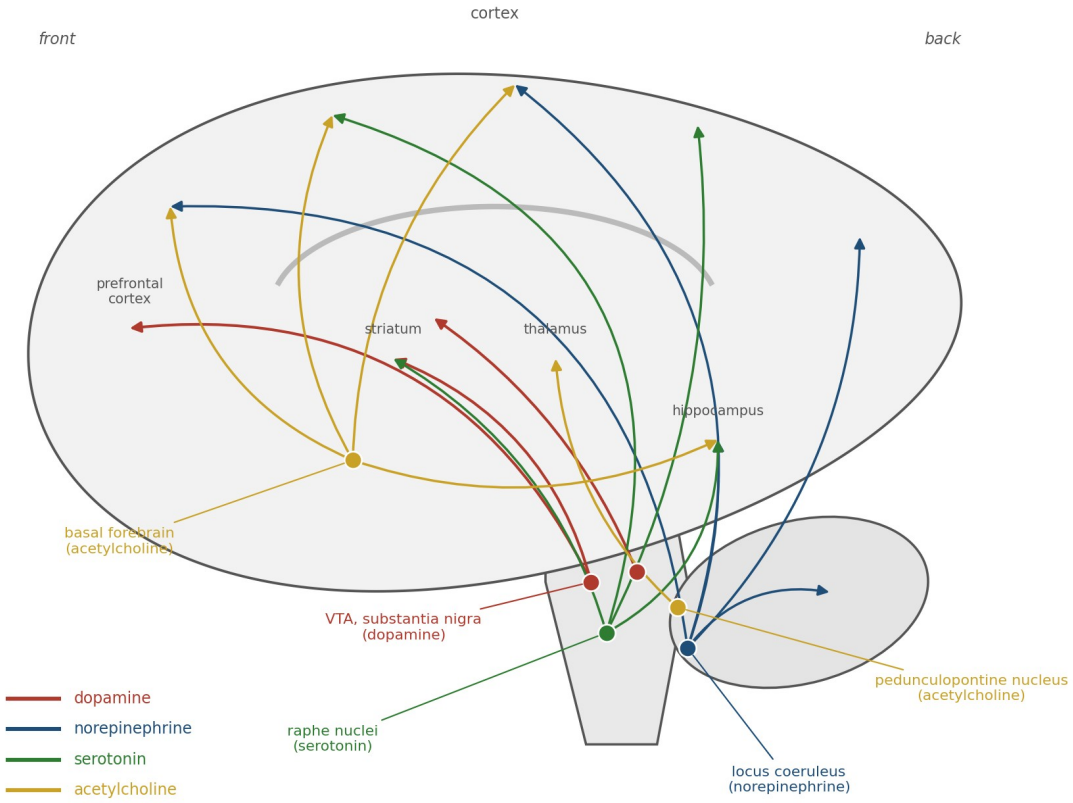


Figure 3.3. Major neuromodulatory systems in the brain, on a schematic midsagittal view with the front at the left. Small nuclei in the brainstem and basal forebrain send dopamine, norepinephrine, serotonin, and acetylcholine across cortex, striatum, thalamus, hippocampus, and cerebellum.

A word on naming. Neuroscientists label neurons by the transmitter they release, with the suffix "-ergic," from the Greek *ergon*, work: dopaminergic neurons release dopamine, noradrenergic neurons

[7] Dahlström, A., & Fuxe, K. (1964). Evidence for the existence of monoamine-containing neurons in the central nervous system. I. Demonstration of monoamines in the cell bodies of brain stem neurons. *Acta Physiologica Scandinavica*, 62, Supplement 232, 1–55; Mesulam, M.-M., Mufson, E. J., Wainer, B. H., & Levey, A. I. (1983). Central cholinergic pathways in the rat: An overview based on an alternative nomenclature (Ch1–Ch6). *Neuroscience*, 10(4), 1185–1201.

release norepinephrine, serotonergic neurons release serotonin, and cholinergic neurons release acetylcholine. The label says what chemical influence a cell exerts, not what it represents.

The numbers are striking. The human nucleus basalis of Meynert, the main cholinergic source for the neocortex, holds roughly 200,000 cholinergic neurons per hemisphere, and the whole basal forebrain cholinergic system amounts to a few hundred thousand cells; the cortex they innervate contains about 16 billion neurons.[8] Figure 3.4 shows that system on its own, and the section on acetylcholine below returns to what it does. A single neuromodulatory neuron compensates with the size of its axon. When Wakoto Matsuda and colleagues filled individual dopamine neurons of the rat substantia nigra and reconstructed their axons, each neuron's arbor in the striatum was about half a meter in total length and covered on average 2.7 percent of the striatum's volume, up to about 6 percent in the largest; estimates of the number of release sites on one such neuron run to several hundred thousand in the rat and more than a million in the human.[9] One cell, in other words, can speak to hundreds of thousands of synapses at once.

[8] Mesulam, M.-M., & Geula, C. (1988). Nucleus basalis (Ch4) and cortical cholinergic innervation in the human brain: Observations based on the distribution of acetylcholinesterase and choline acetyltransferase. *Journal of Comparative Neurology*, 275(2), 216–240; Azevedo, F. A. C., Carvalho, L. R. B., Grinberg, L. T., Farfel, J. M., Ferretti, R. E. L., Leite, R. E. P., Jacob Filho, W., Lent, R., & Herculano-Houzel, S. (2009). Equal numbers of neuronal and nonneuronal cells make the human brain an isometrically scaled-up primate brain. *Journal of Comparative Neurology*, 513(5), 532–541. Mesulam and Geula counted about 210,000 cholinergic neurons in the human nucleus basalis per hemisphere; Azevedo and colleagues put the human cerebral cortex at about 16 billion neurons.

[9] Matsuda, W., Furuta, T., Nakamura, K. C., Hioki, H., Fujiyama, F., Arai, R., & Kaneko, T. (2009). Single nigrostriatal dopaminergic neurons form widely spread and highly dense axonal arborizations in the neostriatum. *Journal of Neuroscience*, 29(2), 444–453; Bolam, J. P., & Pissadaki, E. K. (2012). Living on the edge with too many mouths to feed: Why dopamine neurons die. *Movement Disorders*, 27(12), 1478–1483. Matsuda and colleagues report arbors covering on average 2.7 percent of the rat neostriatum (range about 0.5 to 5.7 percent); Bolam and Pissadaki estimate several hundred thousand release sites per rat neuron and more than a million per human neuron.

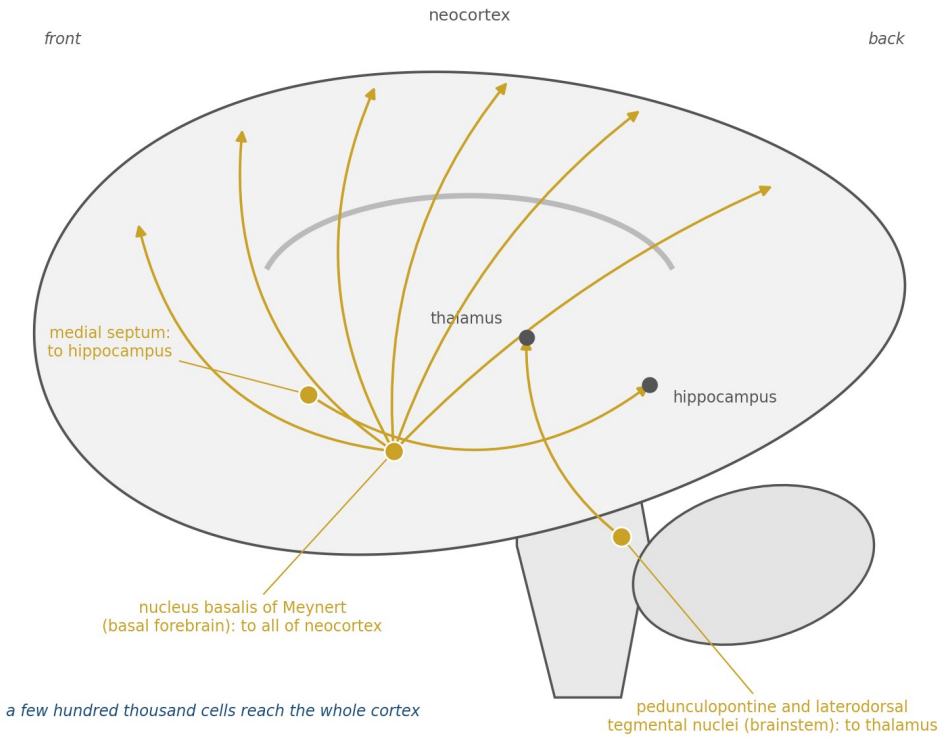


Figure 3.4. The cholinergic system. The nucleus basalis of Meynert reaches all of neocortex, the medial septum reaches the hippocampus, and the pedunculopontine and laterodorsal tegmental nuclei of the brainstem reach the thalamus. A few hundred thousand cells innervate a cortex of about 16 billion neurons.

Why broadcast a signal so widely rather than deliver it to the circuits that need it? Because what these systems carry is context, not instruction. They report whether the organism is safe or threatened, whether events are going better or worse than expected, and whether the environment is stable or has just changed, and those are facts that every circuit needs at the same time.[10] A broadcast lets a few hundred thousand cells coordinate the behavior of billions without a central controller deciding who should hear.[11]

Dopamine: Schultz's Monkeys

Dopamine is often described as the brain's reward or pleasure chemical, and the description is wrong in an instructive way.[12] What corrected it was a series of experiments by Wolfram Schultz and colleagues, beginning in the 1980s and summarized in a 1998 review, in which fine microelectrodes recorded the

[10] Yu, A. J., & Dayan, P. (2005). Uncertainty, neuromodulation, and attention. *Neuron*, 46(4), 681–692; Doya, K. (2002). Metalearning and neuromodulation. *Neural Networks*, 15(4–6), 495–506; Dayan, P. (2012). Twenty-five lessons from computational neuromodulation. *Neuron*, 76(1), 240–256.

[11] Lee, S.-H., & Dan, Y. (2012). Neuromodulation of brain states. *Neuron*, 76(1), 209–222.

[12] Berridge, K. C., & Robinson, T. E. (1998). What is the role of dopamine in reward: Hedonic impact, reward learning, or incentive salience? *Brain Research Reviews*, 28(3), 309–369; Wise, R. A. (2004). Dopamine, learning and motivation. *Nature Reviews Neuroscience*, 5(6), 483–494.

activity of single dopamine neurons in the midbrain of macaque monkeys while the monkeys performed simple tasks for drops of fruit juice.[13]

The neurons fired at a slow, steady rate, a few spikes per second, most of the time. When a drop of juice arrived unannounced, they produced a brief burst, starting within about a tenth of a second and lasting about as long. So far this looked like a reward signal. The monkeys were then trained with a cue, a light or a sound that preceded the juice by a second or so. Over days, as the monkeys learned that the cue predicted the juice, the burst moved. It now came at the cue, and the juice itself, when it arrived as predicted, evoked nothing at all. Finally, on trials in which the cue appeared but the juice was withheld, the neurons did something a reward detector cannot do: at the moment the juice should have arrived, they fell silent, dipping below their baseline rate for a few hundred milliseconds. The dip was timed to the expected delivery, which showed that the neurons carried a prediction of when, not merely whether.[14] Figure 3.5 shows the three conditions as simulated rasters and histograms.

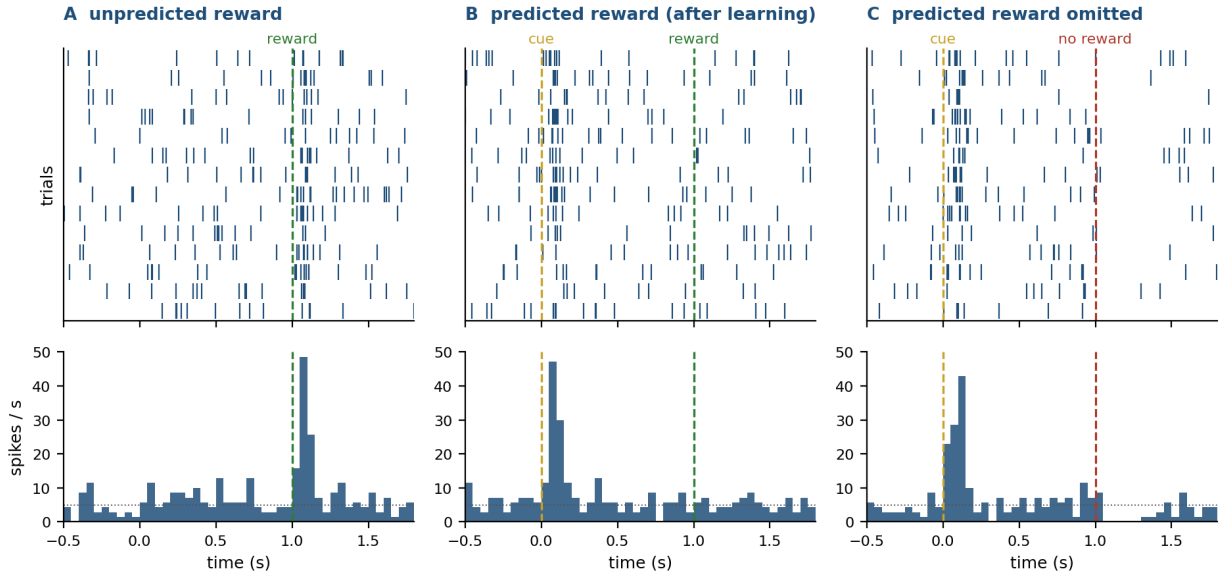


Figure 3.5. Dopamine neurons signal a reward prediction error. Simulated spike rasters and firing-rate histograms in the three conditions of Schultz's experiments. An unpredicted reward evokes a burst (A). After learning, the burst moves to the cue and the predicted reward evokes nothing (B). When a predicted reward is withheld, firing dips below baseline at the moment the reward was due (C).

The neurons were not reporting reward. They were reporting the difference between the reward the monkey received and the reward it had predicted. An unpredicted reward is a positive difference: a burst. A fully predicted reward is no difference: silence. A predicted reward that fails to arrive is a negative difference: a dip. This quantity is called a reward prediction error, and later work showed that

[13] Schultz, W., Apicella, P., & Ljungberg, T. (1993). Responses of monkey dopamine neurons to reward and conditioned stimuli during successive steps of learning a delayed response task. *Journal of Neuroscience*, 13(3), 900–913; Schultz, W. (1998). Predictive reward signal of dopamine neurons. *Journal of Neurophysiology*, 80(1), 1–27.

[14] Hollerman, J. R., & Schultz, W. (1998). Dopamine neurons report an error in the temporal prediction of reward during learning. *Nature Neuroscience*, 1(4), 304–309; Schultz, W. (1998). Predictive reward signal of dopamine neurons. *Journal of Neurophysiology*, 80(1), 1–27.

the size of the burst scales with the size of the error, so that the signal is quantitative rather than merely present or absent.[15]

The interpretation that made these recordings famous came from computer science. In 1988 Richard Sutton had described an algorithm for learning to predict future reward called temporal-difference learning. At each moment the learner holds an estimate of how much reward is still to come. When the next moment arrives it compares the reward actually received, plus its new estimate of what remains, against the estimate it held a moment before; the difference between these two successive predictions, the temporal difference, is the error it uses to correct the earlier one. Over repeated trials the correction propagates backward in time, from the reward to whatever reliably preceded it.[16] In 1996 Read Montague, Peter Dayan, and Terrence Sejnowski observed that Schultz's recordings matched this error term in detail, and in 1997 Schultz, Dayan, and Montague laid the two side by side: the burst that migrates from the juice to the cue, and the dip when the predicted juice is missing, are what a temporal-difference error looks like as learning proceeds.[17] Figure 3.6 shows the migration in a simulation of the algorithm. Within a trial the error starts at the reward and, over trials, moves back to the cue, so that the error at the reward falls to zero as the error at the cue rises.

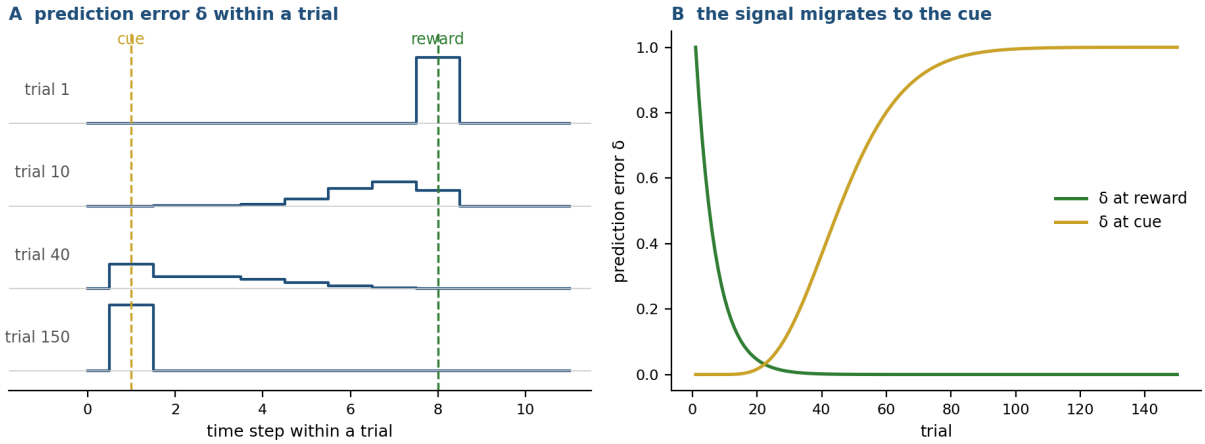


Figure 3.6. Dopamine signals shift to predictive cues after learning. In a simulation of temporal-difference learning, the prediction error within a trial starts at the reward and, over trials, migrates backward to the cue (A), so the error at the reward falls to zero while the error at the cue rises to the full reward value in this undiscounted simulation (B).

A correlation between a neuron's firing and an algorithm's variable is not proof that the neuron causes learning. The causal test came later. When dopamine neurons in rats were driven with light, by

[15] Bayer, H. M., & Glimcher, P. W. (2005). Midbrain dopamine neurons encode a quantitative reward prediction error signal. *Neuron*, 47(1), 129–141; Eshel, N., Bukwich, M., Rao, V., Hemmelder, V., Tian, J., & Uchida, N. (2015). Arithmetic and local circuitry underlying dopamine prediction errors. *Nature*, 525(7568), 243–246.

[16] Sutton, R. S. (1988). Learning to predict by the methods of temporal differences. *Machine Learning*, 3(1), 9–44; Sutton, R. S., & Barto, A. G. (2018). *Reinforcement Learning: An Introduction* (2nd ed.). MIT Press; Rescorla, R. A., & Wagner, A. R. (1972). A theory of Pavlovian conditioning: Variations in the effectiveness of reinforcement and nonreinforcement. In A. H. Black & W. F. Prokasy (Eds.), *Classical Conditioning II: Current Research and Theory* (pp. 64–99). Appleton-Century-Crofts. The Rescorla-Wagner model is the trial-level ancestor of the temporal-difference rule.

[17] Montague, P. R., Dayan, P., & Sejnowski, T. J. (1996). A framework for mesencephalic dopamine systems based on predictive Hebbian learning. *Journal of Neuroscience*, 16(5), 1936–1947; Schultz, W., Dayan, P., & Montague, P. R. (1997). A neural substrate of prediction and reward. *Science*, 275(5306), 1593–1599.

optogenetics, at the moment a reward would normally have produced no error because it was already predicted, the rats learned about a cue they would otherwise have ignored, as if an error had occurred.[18]

Two cautions belong here. First, the reward-prediction-error account describes the phasic response of a large population of midbrain dopamine neurons in tasks with appetitive outcomes; it is not a description of all dopamine neurons or of everything dopamine does. Some dopamine neurons respond to aversive events as well as rewarding ones, some to novelty or salience, and recordings from dopamine neurons and axons in mice have found signals tied to movement, position, and other variables; whether the slow ramps of dopamine seen in the striatum are prediction errors or something else is an active debate.[19] The right statement is that a large population of dopamine neurons signals a reward prediction error, and that the population is heterogeneous in ways still being worked out. Second, the error in question is a single number about value, about how much better or worse things went than expected. Chapter 6 introduces a different prediction error, the difference between what the senses deliver and what the cortex expected them to deliver, which is high-dimensional and about content rather than value. The two share a name and a logic, but they are different signals.

What the dopamine signal does not do is say which synapses should change. It is broadcast to the entire striatum and much of the cortex at once, and it carries one scalar. It does say when, and in which direction: bursts drive potentiation at synapses onto striatal neurons that carry one class of dopamine receptor, while dips favor the opposite change at synapses carrying the other, and the burst must arrive within a narrow window after the synapse was active, a point taken up below.[20]

Credit Assignment

Underneath all learning lies a question that Marvin Minsky named in 1961: when an action leads to success or failure, which of the many internal components that contributed deserve the credit or the blame? This is the credit assignment problem.[21]

Artificial neural networks solve it with care. The architecture is known, the objective is explicit, and the error at the output can be computed by comparison with a correct answer. The backpropagation algorithm then carries that error backward through the network, layer by layer, and assigns to every

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[18] Steinberg, E. E., Keiflin, R., Boivin, J. R., Witten, I. B., Deisseroth, K., & Janak, P. H. (2013). A causal link between prediction errors, dopamine neurons and learning. *Nature Neuroscience*, 16(7), 966–973.

[19] Matsumoto, M., & Hikosaka, O. (2009). Two types of dopamine neuron distinctly convey positive and negative motivational signals. *Nature*, 459(7248), 837–841; Bromberg-Martin, E. S., Matsumoto, M., & Hikosaka, O. (2010). Dopamine in motivational control: Rewarding, aversive, and alerting. *Neuron*, 68(5), 815–834; Howe, M. W., & Dombeck, D. A. (2016). Rapid signalling in distinct dopaminergic axons during locomotion and reward. *Nature*, 535(7613), 505–510; Engelhard, B., Finkelstein, J., Cox, J., Fleming, W., Jang, H. J., Ornelas, S., Koay, S. A., Thiberge, S. Y., Daw, N. D., Tank, D. W., & Witten, I. B. (2019). Specialized coding of sensory, motor and cognitive variables in VTA dopamine neurons. *Nature*, 570(7762), 509–513; Jeong, H., Taylor, A., Floeder, J. R., Lohmann, M., Mihalas, S., Wu, B., Zhou, M., Burke, D. A., & Namboodiri, V. M. K. (2022). Mesolimbic dopamine release conveys causal associations. *Science*, 378(6626), eabq6740.

[20] Shen, W., Flajolet, M., Greengard, P., & Surmeier, D. J. (2008). Dichotomous dopaminergic control of striatal synaptic plasticity. *Science*, 321(5890), 848–851; Reynolds, J. N. J., Hyland, B. I., & Wickens, J. R. (2001). A cellular mechanism of reward-related learning. *Nature*, 413(6851), 67–70; Yagishita, S., Hayashi-Takagi, A., Ellis-Davies, G. C. R., Urakubo, H., Ishii, S., & Kasai, H. (2014). A critical time window for dopamine actions on the structural plasticity of dendritic spines. *Science*, 345(6204), 1616–1620. Shen and colleagues showed that D1-receptor and D2-receptor striatal neurons respond to dopamine with opposite directions of plasticity.

[21] Minsky, M. (1961). Steps toward artificial intelligence. *Proceedings of the IRE*, 49(1), 8–30.

weight an update proportional to its exact contribution to the error.[22] It is precise, and it requires a complete, labeled, differentiable description of the system in which it runs.

The brain has no known way to do this exactly, and whether it approximates it is one of the field's open questions. Francis Crick raised the objection in 1989: nothing in the cortex looked like a mechanism for sending each synapse its share of a global error.[23] Since then the objection has weakened without disappearing. Random feedback weights turn out to be enough for a network to learn, so the backward pathway need not mirror the forward one; bursts of spikes in cortical pyramidal cells, driven by inputs to their apical dendrites, can carry an error signal separate from the signal the cell forwards; and Timothy Lillicrap and colleagues argued in 2020 that the brain may well implement something functionally like backpropagation, with the feedback connections that cortical anatomy supplies in abundance carrying the errors.[24] Chapters 5 and 6 return to the question. The point for this chapter is narrower: whatever the brain does about credit assignment, it is not the clean procedure of the artificial case, and a large part of what it does instead is visible in the neuromodulatory systems.

Consider what makes the problem hard. The brain receives no labels. It is never told the correct answer, only whether things went well or badly, and that news arrives late, after a delay of seconds or minutes during which millions of synapses have been active for unrelated reasons. Circuits are recurrent and overlapping rather than neatly layered. Neurons take part in many functions at once. By the time an outcome is known, the detailed pattern of activity that produced it has been overwritten by everything that happened since, a difficulty Eugene Izhikevich called the distal reward problem.[25] In many cases the question "which synapse caused this outcome?" has no single answer, because thousands contributed and none was decisive.

Learning Under Uncertainty

A system that cannot identify exact causes must learn under uncertainty. The brain does not know which synapse mattered or which action was decisive. What it can know is that something important just happened, and the neuromodulatory systems are built to say so.

The approximate solution works in two steps. When an outcome is better or worse than expected, dopamine broadcasts a signal that marks the recent past as significant. Synapses that were active shortly before the signal, and only those, become eligible for modification.[26] No individual update is guaranteed to be correct; a synapse that was active by coincidence is strengthened along with the ones

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[22] Rumelhart, D. E., Hinton, G. E., & Williams, R. J. (1986). Learning representations by back-propagating errors. *Nature*, 323(6088), 533–536.

[23] Crick, F. (1989). The recent excitement about neural networks. *Nature*, 337(6203), 129–132.

[24] Lillicrap, T. P., Cownden, D., Tweed, D. B., & Akerman, C. J. (2016). Random synaptic feedback weights support error backpropagation for deep learning. *Nature Communications*, 7, 13276; Payeur, A., Guerguiev, J., Zenke, F., Richards, B. A., & Naud, R. (2021). Burst-dependent synaptic plasticity can coordinate learning in hierarchical circuits. *Nature Neuroscience*, 24(7), 1010–1019; Lillicrap, T. P., Santoro, A., Marris, L., Akerman, C. J., & Hinton, G. (2020). Backpropagation and the brain. *Nature Reviews Neuroscience*, 21(6), 335–346.

[25] Izhikevich, E. M. (2007). Solving the distal reward problem through linkage of STDP and dopamine signaling. *Cerebral Cortex*, 17(10), 2443–2452.

[26] Frémaux, N., & Gerstner, W. (2016). Neuromodulated spike-timing-dependent plasticity, and theory of three-factor learning rules. *Frontiers in Neural Circuits*, 9, 85; Gerstner, W., Lehmann, M., Liakoni, V., Corneil, D., & Brea, J. (2018). Eligibility traces and plasticity on behavioral time scales: Experimental support of neoHebbian three-factor learning rules. *Frontiers in Neural Circuits*, 12, 53.

that mattered. But the updates are small, and they accumulate. Over many trials, patterns of activity that tend to precede good outcomes are strengthened more often than they are weakened, and patterns that precede bad ones the reverse, which is Edward Thorndike's law of effect from 1911, now with a mechanism.[27] The system acquires biases, statistical tendencies toward what has worked. Bias here does not mean error. It means the gradual shaping of probability under uncertainty, and it is what learning looks like when the learner cannot see the causal chain.

It is tempting to summarize this by saying that the brain, unlike an artificial network, does not optimize a loss function. That is true in one sense and false in another, and the book needs both. No single global objective is handed to the brain and minimized by gradient descent; there is no external label, no fixed architecture, and no one error. But many circuits are well described as minimizing a local error or cost. The temporal-difference error of Figure 3.6 is such a quantity, and the dopamine system is, to a good approximation, the machinery that reduces it. Chapter 6 describes cortex as reducing a sensory prediction error, and shows the brain combining cues in a way that matches a computed optimum. The accurate statement is that the brain has many local objectives and no global one, and that neuromodulators are among the signals that define them.

Eligibility Traces and the Three-Factor Rule

The two-step solution requires something at the synapse that remembers recent activity long enough for the modulatory signal to arrive. That something is called an eligibility trace. The term comes from reinforcement learning, where it names a decaying record of which states and actions were recently visited; Harry Klopf proposed a synaptic version in 1982, before there was evidence for one.[28]

The evidence now exists. When a synapse is active, with the presynaptic neuron firing shortly before the postsynaptic one, biochemical processes put it into a transient, modifiable state. If a neuromodulatory signal arrives while that state persists, the synapse changes; if not, the state fades and nothing lasting happens. The windows have been measured. In 2014 Sho Yagishita and colleagues, working in slices of mouse striatum, activated a single dendritic spine with a flash of light that released glutamate at it while the neuron fired, then released dopamine at varying delays by stimulating dopamine axons with light. The spine enlarged, which is the structural signature of potentiation, only when dopamine arrived between about 0.3 and 2 seconds after the activity; earlier or later, nothing.[29] In hippocampal slices the window is longer. Zuzanna Brzosko, Wolfram Schultz, and Ole Paulsen found that a pairing protocol that normally produces depression was converted into potentiation if dopamine was applied within about a minute afterward, but not if the delay was ten minutes, and recordings in living rats have found windows of similar length at synapses from cortex onto the striatum.[30] Eligibility traces in the brain,

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[28] Klopf, A. H. (1982). *The Hedonistic Neuron: A Theory of Memory, Learning, and Intelligence*. Hemisphere; Sutton, R. S., & Barto, A. G. (2018). *Reinforcement Learning: An Introduction* (2nd ed.). MIT Press.

[29] Yagishita, S., Hayashi-Takagi, A., Ellis-Davies, G. C. R., Urakubo, H., Ishii, S., & Kasai, H. (2014). A critical time window for dopamine actions on the structural plasticity of dendritic spines. *Science*, 345(6204), 1616–1620.

[30] Brzosko, Z., Schultz, W., & Paulsen, O. (2015). Retroactive modulation of spike timing-dependent plasticity by dopamine. *eLife*, 4, e09685; Fisher, S. D., Robertson, P. B., Black, M. J., Redgrave, P., Sagar, M. A., Abraham, W. C., & Reynolds, J. N. J. (2017). Reinforcement determines the timing dependence of corticostriatal synaptic plasticity in vivo. *Nature Communications*, 8, 334; He, K., Huertas, M., Hong, S. Z., Tie, X., Hell, J. W., Shouval, H., & Kirkwood, A. (2015). Distinct eligibility traces for LTP and LTD in cortical synapses. *Neuron*, 88(3), 528–538. Measured windows run from about a second for dopamine action on striatal

then, last seconds to minutes, which is the timescale on which outcomes follow actions. Figure 3.7 shows the logic: a spike pair sets a trace that decays, a dopamine pulse that arrives during the decay changes the weight, and the same pulse arriving after the trace has gone changes nothing.

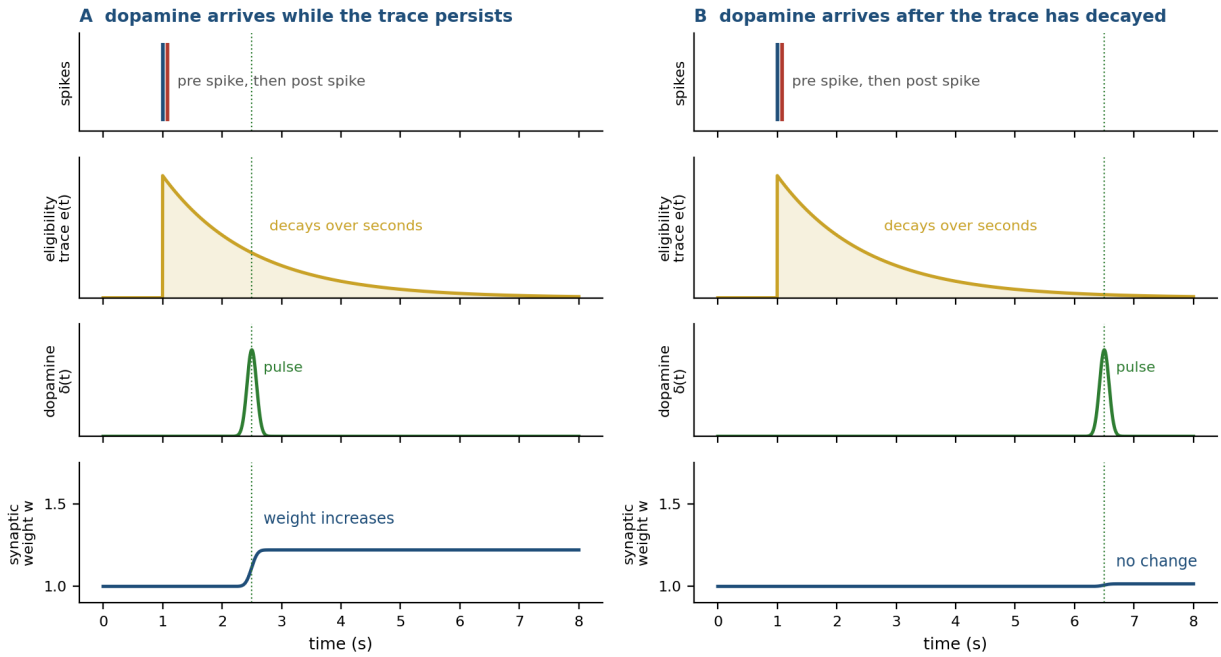


Figure 3.7. Eligibility traces bridge the gap between activity and the learning signal. In this illustrative simulation a pre-then-post spike pair sets a trace that decays over seconds. A dopamine pulse that arrives while the trace persists changes the synaptic weight (A). The same pulse arriving after the trace has decayed changes nothing (B).

Put together, this is a three-factor learning rule. A synapse changes only when three things coincide within the lifetime of the trace: the presynaptic neuron was active, the postsynaptic neuron fired, and a neuromodulatory signal arrived.[31] The first two factors are Hebb's, and Chapter 5 takes them apart; the third is the subject of this chapter. Figure 3.8 writes the rule out. The eligibility trace rises with each spike pairing by an amount set by the relative timing of the two spikes (a rule Chapter 5 explains), decays with a time constant of seconds, and is multiplied by the dopamine signal; the weight moves only while both the trace and the signal are present, and the sign of the pairing and the sign of the signal together set the direction.

spines to about a minute in hippocampal slices; He and colleagues found separate traces for potentiation and depression in cortex, each lasting seconds.

[31] Frémaux, N., & Gerstner, W. (2016). Neuromodulated spike-timing-dependent plasticity, and theory of three-factor learning rules. *Frontiers in Neural Circuits*, 9, 85; Gerstner, W., Lehmann, M., Liakoni, V., Corneil, D., & Brea, J. (2018). Eligibility traces and plasticity on behavioral time scales: Experimental support of neoHebbian three-factor learning rules. *Frontiers in Neural Circuits*, 12, 53.

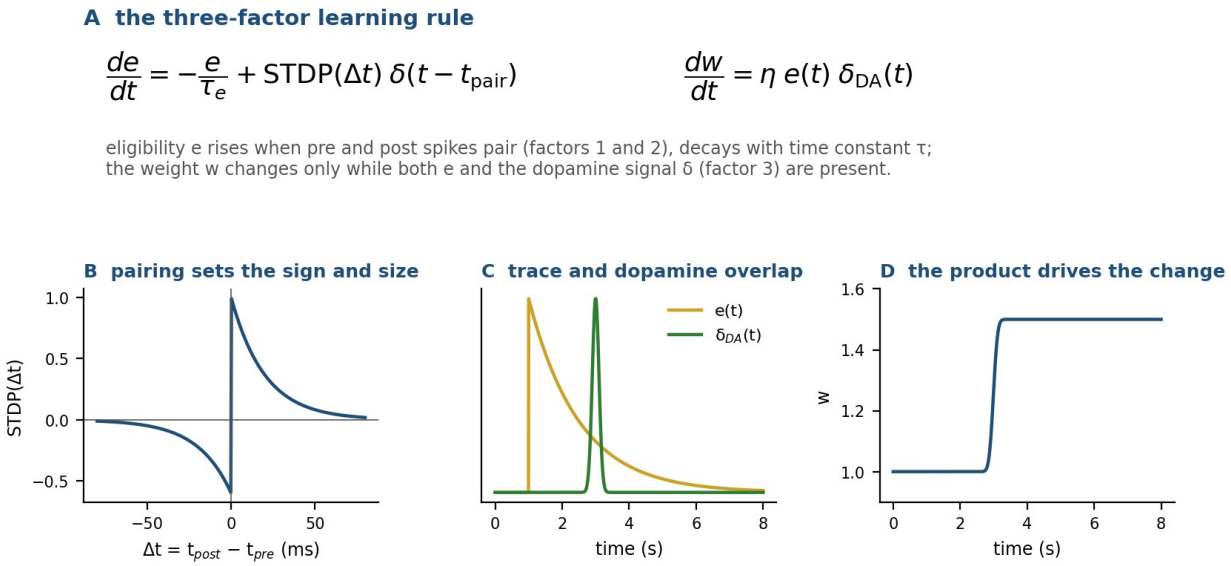


Figure 3.8. The three-factor rule written out (A). The eligibility trace e rises with each spike pairing according to the spike-timing window (B), decays, and is multiplied by the dopamine signal (C); the weight changes only while both are present (D). The curves are illustrative.

Eligibility traces let learning bridge the gap between action and outcome without any knowledge of the causal chain, and they are the reason a scalar broadcast can produce selective change: the signal reaches every synapse, but only the recently active ones are listening. What physically happens when a trace is captured, the receptors inserted, the spine enlarged, the proteins synthesized, is the subject of Chapter 5, which also describes synaptic tagging and capture, the molecular form of the trace, in which a tag set at a synapse lets it collect proteins whose synthesis a modulatory signal triggers up to an hour or two later.[32]

Norepinephrine, Serotonin, and Acetylcholine

Dopamine is the best understood of the four systems because reward is the easiest context to manipulate in an experiment. The other three are at least as important, and each has been tied to a context of its own.

Norepinephrine, from the locus coeruleus, is the signal of surprise and arousal. Locus coeruleus neurons fire a brief burst at a novel or salient stimulus, and their tonic rate tracks the animal's state from drowsiness through alert engagement to agitation. Gary Aston-Jones and Jonathan Cohen proposed that the system sets the gain of cortical circuits in one of two modes: a phasic mode, in which bursts sharpen responses to task-relevant events, and a tonic mode, in which a high baseline makes the animal responsive to everything and disengaged from the current task, which is the right setting when the current task has stopped paying.[33] In the computational account of Angela Yu and Peter Dayan,

[32] Kandel, E. R. (2001). The molecular biology of memory storage: A dialogue between genes and synapses. *Science*, 294(5544), 1030–1038; Malenka, R. C., & Bear, M. F. (2004). LTP and LTD: An embarrassment of riches. *Neuron*, 44(1), 5–21; Frey, U., & Morris, R. G. M. (1997). Synaptic tagging and long-term potentiation. *Nature*, 385, 533–536.

[33] Aston-Jones, G., & Cohen, J. D. (2005). An integrative theory of locus coeruleus–norepinephrine function: Adaptive gain and optimal performance. *Annual Review of Neuroscience*, 28, 403–450; Sara, S. J. (2009). The locus coeruleus and noradrenergic

norepinephrine reports unexpected uncertainty, the signal that the world has changed and the current model should be distrusted, while acetylcholine reports the expected uncertainty of a world that is noisy but stable.[34] Chapter 6 builds on this distinction when it treats attention as the weighting of evidence by its reliability.

Serotonin, from the raphe nuclei, is the least settled. It has been tied to patience, to aversion, to behavioral inhibition, and to the regulation of plasticity itself, and the reviews that try to unify these accounts are candid that no unification has yet succeeded.[35] One concrete result stands out. Katsuhiko Miyazaki, Kayoko Miyazaki, and Kenji Doya recorded from serotonin neurons of the dorsal raphe while rats waited at a port for a reward that came only after a delay. The neurons raised their firing throughout the wait and dropped it when the rat gave up and left.[36] On one account, then, serotonin promotes the stability that lets an animal hold a course of action against the pull of an immediate alternative, which is roughly the opposite of what dopamine does when it marks a new option as better than expected.

Acetylcholine sets the balance between taking in new information and using old. Its anatomy, shown in Figure 3.4, comes in three parts: the nucleus basalis of Meynert in the basal forebrain sends axons to all of the neocortex, the medial septum reaches the hippocampus, and the pedunculopontine and laterodorsal tegmental nuclei of the brainstem reach the thalamus. Michael Hasselmo and colleagues showed that acetylcholine selectively suppresses transmission at the recurrent synapses by which cortical and hippocampal circuits complete stored patterns, while leaving the inputs that carry new information intact. High acetylcholine, as during attentive waking, therefore favors encoding: the circuit is driven by what is arriving and is prevented from being captured by what it already knows. Low acetylcholine, as during slow-wave sleep, favors recall and the replay of stored patterns that Chapter 5 connects to consolidation.[37] The map-expansion experiment described earlier, in which a tone paired with nucleus basalis stimulation claimed a larger share of auditory cortex, is the same system acting as permission for cortical plasticity.[38]

Together these systems regulate not what the brain thinks but how it thinks: how strongly to respond, how much to trust the current model, whether to persist or to switch, and whether to write or to read.

Clinical Evidence

The clearest evidence that neuromodulatory tone shapes learning and behavior comes from the clinic, where it has been altered deliberately in humans for more than sixty years.

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modulation of cognition. *Nature Reviews Neuroscience*, 10(3), 211–223.

[34] Yu, A. J., & Dayan, P. (2005). Uncertainty, neuromodulation, and attention. *Neuron*, 46(4), 681–692.

[35] Dayan, P., & Huys, Q. J. M. (2009). Serotonin in affective control. *Annual Review of Neuroscience*, 32, 95–126; Cools, R., Roberts, A. C., & Robbins, T. W. (2008). Serotoninergic regulation of emotional and behavioural control processes. *Trends in Cognitive Sciences*, 12(1), 31–40.

[36] Miyazaki, K., Miyazaki, K. W., & Doya, K. (2011). Activation of dorsal raphe serotonin neurons underlies waiting for delayed rewards. *Journal of Neuroscience*, 31(2), 469–479.

[37] Hasselmo, M. E., & Bower, J. M. (1993). Acetylcholine and memory. *Trends in Neurosciences*, 16(6), 218–222; Hasselmo, M. E. (2006). The role of acetylcholine in learning and memory. *Current Opinion in Neurobiology*, 16(6), 710–715.

[38] Kilgard, M. P., & Merzenich, M. M. (1998). Cortical map reorganization enabled by nucleus basalis activity. *Science*, 279(5357), 1714–1718.

Parkinson's disease is the founding case. In 1960 Herbert Ehringer and Oleh Hornykiewicz measured dopamine in postmortem brains and found it almost absent from the striatum of patients who had died with the disease, the first time a neurological disorder had been tied to the loss of a specific transmitter.[39] Arvid Carlsson had shown three years earlier that animals immobilized by reserpine, a drug that empties the brain's stores of monoamines, recovered when given L-DOPA, the precursor from which neurons make dopamine; in 1967 George Cotzias gave patients L-DOPA by mouth in gradually increasing doses and reported, in the sober prose of a medical journal, that rigidity and slowness of movement were in some patients nearly abolished.[40] L-DOPA remains the standard treatment. Its effect on learning, as distinct from movement, was shown by Michael Frank, Lauren Seeberger, and Randall O'Reilly in 2004. Patients chose repeatedly between pairs of symbols that paid off with different probabilities. Off medication, with striatal dopamine low, they were better at learning which symbol to avoid than which to choose; on medication, the reverse. The same patients, the same task, and a shift in dopamine tone changed whether they learned more from good outcomes or from bad.[41] This is what the three-factor rule predicts if bursts and dips carry different lessons and medication raises the floor under the dips.

Dopamine can also be reduced. The antipsychotic drugs used for schizophrenia block dopamine receptors, and in 1976 Ian Creese, David Burt, and Solomon Snyder showed that the clinical dose of each drug was predicted by how tightly it bound the dopamine receptor that would later be called D2; the drugs' parkinsonian side effects, slowed movement and tremor, are dopamine loss produced by pharmacology.[42] Acetylcholine has a parallel story. In 1982 Peter Whitehouse and colleagues found that the nucleus basalis of patients who had died with Alzheimer's disease had lost most of its neurons, and the cholinergic hypothesis of the memory loss of dementia followed. The drugs built on it, donepezil, galantamine, and rivastigmine, inhibit the enzyme that breaks acetylcholine down and so raise its level at every cholinergic synapse. A Cochrane review of the trials found that they help: a modest but consistent improvement in cognitive scores at six months, which is what one would expect from raising the tone of a system that permits encoding without restoring the neurons that were lost.[43]

Deep brain stimulation belongs in this section with a caveat attached. In 1991 Alim-Louis Benabid and colleagues reported that chronic electrical stimulation of a thalamic nucleus through an implanted electrode suppressed tremor, and in 1998 Patricia Limousin and colleagues extended the method to the subthalamic nucleus in advanced Parkinson's disease, where it relieves rigidity and slowness and lets medication be reduced.[44] Figure 3.9 shows the hardware: a lead through a burr hole in the skull, its tip in

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[39] Ehringer, H., & Hornykiewicz, O. (1960). Verteilung von Noradrenalin und Dopamin (3-Hydroxytyramin) im Gehirn des Menschen und ihr Verhalten bei Erkrankungen des extrapyramidalen Systems. *Klinische Wochenschrift*, 38, 1236–1239.

[40] Carlsson, A., Lindqvist, M., & Magnusson, T. (1957). 3,4-Dihydroxyphenylalanine and 5-hydroxytryptophan as reserpine antagonists. *Nature*, 180(4596), 1200; Cotzias, G. C., Van Woert, M. H., & Schiffer, L. M. (1967). Aromatic amino acids and modification of parkinsonism. *New England Journal of Medicine*, 276(7), 374–379.

[41] Frank, M. J., Seeberger, L. C., & O'Reilly, R. C. (2004). By carrot or by stick: Cognitive reinforcement learning in parkinsonism. *Science*, 306(5703), 1940–1943.

[42] Creese, I., Burt, D. R., & Snyder, S. H. (1976). Dopamine receptor binding predicts clinical and pharmacological potencies of antischizophrenic drugs. *Science*, 192(4238), 481–483.

[43] Whitehouse, P. J., Price, D. L., Struble, R. G., Clark, A. W., Coyle, J. T., & DeLong, M. R. (1982). Alzheimer's disease and senile dementia: Loss of neurons in the basal forebrain. *Science*, 215(4537), 1237–1239; Birks, J. S. (2006). Cholinesterase inhibitors for Alzheimer's disease. *Cochrane Database of Systematic Reviews*, (1), CD005593.

[44] Benabid, A. L., Pollak, P., Gervason, C., Hoffmann, D., Gao, D. M., Hommel, M., Perret, J. E., & de Rougemont, J. (1991). Long-term suppression of tremor by chronic stimulation of the ventral intermediate thalamic nucleus. *The Lancet*, 337(8738), 403–406; Limousin, P., Krack, P., Pollak, P., Benazzouz, A., Ardouin, C., Hoffmann, D., & Benabid, A.-L. (1998). Electrical stimulation of

a nucleus a few millimeters across, connected under the skin to a pulse generator below the collarbone. The procedure is often called neuromodulation, and in the clinical sense of the word it is. In the sense used in this chapter it is not. The stimulation is electrical, it acts on the activity of a circuit rather than on the level of any transmitter, and how it works, whether by silencing the target, by driving it in a regular rhythm that the rest of the circuit can ignore, or by disrupting a pathological oscillation, is still debated.[45] It is strong evidence that the basal ganglia circuit of Chapter 4 can be pushed from outside. It is not direct evidence about neuromodulator tone, and it is placed here so that the two senses of the word are not confused.

the subthalamic nucleus in advanced Parkinson's disease. *New England Journal of Medicine*, 339(16), 1105–1111.

[45] Lozano, A. M., Lipsman, N., Bergman, H., Brown, P., Chabardes, S., Chang, J. W., Matthews, K., McIntyre, C. C., Schlaepfer, T. E., Schulder, M., Temel, Y., Volkmann, J., & Krauss, J. K. (2019). Deep brain stimulation: Current challenges and future directions. *Nature Reviews Neurology*, 15(3), 148–160.

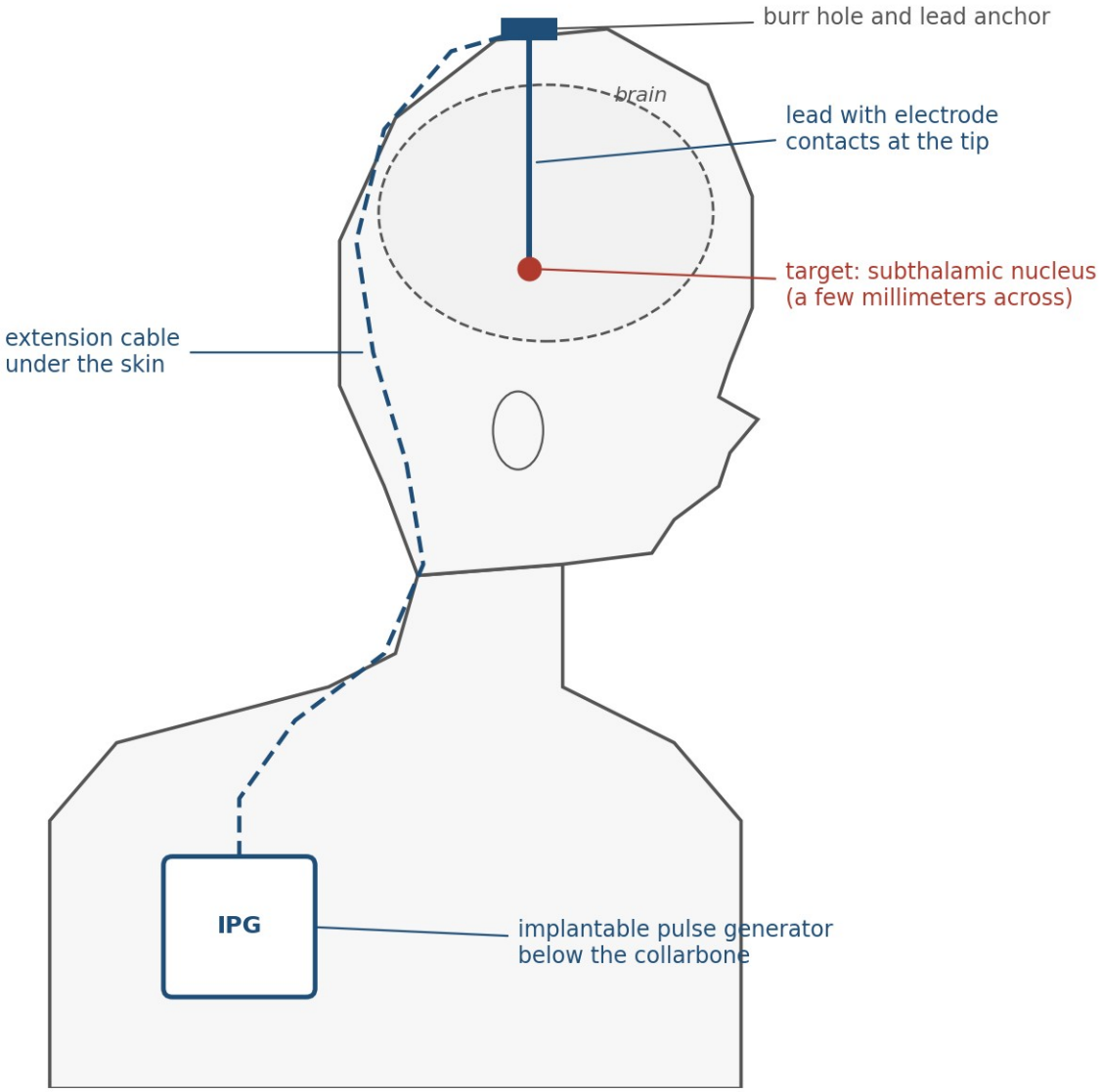


Figure 3.9. Deep brain stimulation hardware in place. A lead enters through a burr hole in the skull and ends in the subthalamic nucleus; an extension cable under the skin connects it to a pulse generator implanted below the collarbone. The intervention is electrical and acts on circuit activity, not on neuromodulator levels.

Implications for Artificial Intelligence

Modern artificial neural networks are extraordinarily capable within the assumptions they were built on. They learn well when the objective is explicit, feedback is immediate, and the architecture is fixed,

and under those conditions backpropagation gives each weight an exact gradient, which is possible precisely because the conditions hold. Biological intelligence evolved under none of them. Real organisms learn where goals are ambiguous, feedback is delayed, and the causal structure of outcomes is hidden. The brain's mechanisms reflect that: a global modulatory signal, local eligibility traces, and incremental biasing of behavior over time rather than an exact gradient. Several implications follow.

First, the brain separates activity from learning, and does so from the inside. In an artificial system the separation is imposed from outside by a switch. During training every forward pass is followed by a weight update; in deployment the weights are frozen and no forward pass changes anything. The brain has no such switch. It runs and learns continuously, and the decision about whether a given moment of activity is allowed to change the wiring is made internally, by the neuromodulatory context, on a timescale of seconds. Nothing changes unless the context says it should. A system that must keep learning after deployment needs something of this kind.

Second, learning rules themselves can be state-dependent. Neuromodulators change how the same activity is interpreted: what counts as a surprise, how large an update is warranted, whether to encode or to recall. Kenji Doya proposed in 2002 that the four systems correspond to the four quantities a reinforcement learner has to set, the error signal, the learning rate, the exploration rate, and the discount on future reward, and recent work has trained networks whose plasticity is gated by a learned modulatory signal, or has found that a recurrent network trained on many tasks develops within its activity a fast learning system that resembles the one dopamine trains in prefrontal cortex.[46]

Third, approximate credit assignment can be enough. A network trained with random feedback weights learns almost as well as one with the exact backward pathway, which means precise attribution is not a requirement for useful adaptation.[47] The brain appears to go further still, broadcasting a single number to an entire region, and it learns well enough to survive.

Finally, neuromodulation reframes the objective. No loss function is handed to the brain. It has many local errors, the temporal-difference error among them, and it reduces them under a constraint that an artificial system rarely faces: it must remain viable while it learns.[48] Success is defined operationally, by survival and by what works, and the neuromodulatory systems are the brain's running estimate of how that is going. As artificial systems move out of curated datasets and into open-ended interaction with the world, these principles may prove more important than further refinements of gradient descent.

The Central Lesson

Learning is gated, broadcast, and biased. The brain has no known way to trace an outcome exactly back to the synapses that caused it, and it does not need one. Instead a few hundred thousand neuromodulatory neurons broadcast a running summary of how things are going, whether events are

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[46] Doya, K. (2002). Metalearning and neuromodulation. *Neural Networks*, 15(4–6), 495–506; Miconi, T., Rawal, A., Clune, J., & Stanley, K. O. (2019). Backpropamine: Training self-modifying neural networks with differentiable neuromodulated plasticity. *International Conference on Learning Representations (ICLR)*; Wang, J. X., Kurth-Nelson, Z., Kumaran, D., Tirumala, D., Soyer, H., Leibo, J. Z., Hassabis, D., & Botvinick, M. (2018). Prefrontal cortex as a meta-reinforcement learning system. *Nature Neuroscience*, 21(6), 860–868.

[47] Lillicrap, T. P., Cownden, D., Tweed, D. B., & Akerman, C. J. (2016). Random synaptic feedback weights support error backpropagation for deep learning. *Nature Communications*, 7, 13276.

[48] Ashby, W. R. (1952). *Design for a Brain*. Chapman & Hall.

better or worse than expected, whether the world has changed, whether to take in or to recall, and that summary decides whether the activity of the moment is permitted to leave a lasting trace. The signal is a scalar that reaches everything, and it becomes selective only because synapses hold a short-lived record of their own recent activity. The changes it produces are small and sometimes wrong, but they accumulate, and over time they bias neural activity and behavior toward the patterns that have tended to succeed.

Next Chapter

Chapter 4: Circuits, Networks, and Coordinated Behavior

References and Attribution

Figure Attribution

Figures 3.1 to 3.9 were drawn for this book. The response traces in Figure 3.2 and the spike rasters, histograms, and curves in Figures 3.5 to 3.8 are illustrative simulations, not recorded data; Figure 3.6 is an undiscounted temporal-difference simulation. The brain outlines in Figures 3.3, 3.4, and 3.9 are schematic, nuclei are placed approximately, and the hardware in Figure 3.9 is generic rather than any manufacturer's design.

Epigraph

Santiago Ramón y Cajal, *Recollections of My Life*, translated by E. Horne Craigie with the assistance of Juan Cano (Memoirs of the American Philosophical Society, vol. 8, Philadelphia, 1937; reprinted MIT Press, 1989), from the chapter on his first work with the Golgi method. The Spanish original, "las misteriosas mariposas del alma, cuyo batir de alas quién sabe si esclarecerá algún día el secreto de la vida mental," is in *Recuerdos de mi vida*, 3rd ed. (Madrid, 1923).

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